MIL-L-36696A
22 September 1978
SUPERSEDING
MIL-L-36696
15 September 1970 and
AMENDMENT 1
1 August 1973

MILITARY SPECIFICATION

LARYNGOSCOPE, CHILD-ADULT, WISCONSIN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a detachable laryngoscope of the Wisconsin design, suitable for use in medical procedures concerning the human larynx.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bid or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

GG-I-526	- Instruments, Dental and Surgical; General Specifications.
PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, Code ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6515

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MILITARY

MIL-P-116 - Preservation-Packaging, Methods of.
* MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-L-36628 - Laryngoscope Set, MacIntosh.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

DEFENSE PERSONNEL SUPPORT CENTER (DPSC)

1249 - Lamp, Incandescent, 2.5 Volt, 0.28 Ampere, Bulb, T-1-1/2, Frosted.
2234 - Laryngoscope, Child, Adult, Wisonsin Type.
* 24276 - Handle, Laryngoscope, Battery Type, Universal, Large.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B1.1 Unified Screw Threads.
ANSI B1.2 Gages and Gaging for Unified Screw Threads.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Tests for.
E 92 - Vickers Hardness of Metallic Materials, Test for.
E 140 - Hardness Conversion Tables for Metals (Relationship Between Brinell Hardness, Vickers Hardness, Rockwell Hardness, Rockwell Superficial Hardness and Knoop Hardness).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Technical society and technical associations and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

3. REQUIREMENTS

3.1 Material.

* 3.1.1 Laryngoscope blade, housing lock, lamp carrier and lamp socket. Shall be fabricated from materials specified in MIL-L-36628. Rockwell hardness of class 3 material shall be B75 minimum when tested as specified in 4.4.2.

The plating and corrision requirements for the alternate brass compositions specified above, shall be in accordance with QQ-I-526 for class 8, type 260 material.

* 3.1.2 Battery handle. Battery handle shall be manufactured from materials specified in MIL-L-36628.

* 3.1.3 Lamps. Lamps shall be manufactured from materials specified in MIL-L-36628.

* 3.2 Style, design and dimensions. The complete laryngoscope shall consist of a corrugated box (see 5.1.2.4) containing all of the following components in the specified quantities:

Component	Size	Quantity
Laryngoscope Blade	Child	1
Laryngoscope Blade	Medium Adult	1
Laryngoscope Blade	Large Adult	1
Handle, Laryngoscope, Battery Type, Universal, Large (When furnished separately, NSN 6515-00-890-2190)	-----	1
Lamp, Incandescent, 2.5 Volt, 0.28 Ampere, Bulb T-1-1/2, Frosted (When furnished separately, NSN 6240-00-797-0420)	Type II Lamp as specified in MIL-L-36628	6 (Includes extra lamp) 1/

1/ Each laryngoscope of the sizes listed above shall be furnished with one lamp assembled to each blade, and one extra lamp for each blade.

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3.2.1 Laryngoscope blade assemblies. The laryngoscope blade assemblies shall be fabricated from material as specified in 3.1.1 and shall be furnished in the sizes and quantities as specified in 3.2. The blade assemblies shall be of the Wisconsin design, and of the detachable and hook-on type. The laryngoscopes shall be suitable for use in positioning tubes into the human larynx. The blades shall be capable of insertion onto the battery handle, which shall be furnished with the set, and designed to light at the distal end when assembled to the battery handle. Each laryngoscope blade shall contain one assembled frosted lamp. Lamps shall be in accordance with MIL-L-36628 for Type II lamps. Three additional spare lamps shall be furnished. The blade assemblies shall be in strict accordance with DPSC Drawing 2234.

3.2.1.1 Internal threads of lamp socket. Internal threads of the lamp socket for each laryngoscope blade shall allow the "GO" thread plug gage to be inserted and turned freely and shall not allow entry of the "HI" and "NOT GO" thread plug gages when tested as specified in 4.4.1. Definitions of acceptance and rejection of the gaging tests shall be as specified in ANSI Standard B1.2.

* 3.2.2. Battery handle. Unit shall be furnished with a large size, universal handle in accordance with MIL-L-36628 and DPSC Drawing 24276.

* 3.2.3 Lamps. Shall be in accordance with MIL-L-36628 for Type II lamps.

* 3.3 Finish. Finish of the laryngoscope assemblies shall be mirror and satin as designated on DPSC Drawing 2234. The battery handle shall have a bright finish and shall be knurled as shown on DPSC Drawing 24276. Finish on all edges and surfaces be uniform, free from burrs, sharp edges, pits, crevices, cracks, rough areas, grind marks, scale or corrosion.

* 3.4 Identification marking. Laryngoscope blade assemblies and battery handle shall be permanently and legibly marked with the contractor's name or registered trademark and country of origin when manufactured in countries outside of the United States. Blade assemblies shall be permanently and legibly marked with the blade name "WISCONSIN" or "WIS" and applicable blade size. Laryngoscope blades manufactured from class 3 material shall also be marked "stainless".

3.6 Workmanship. Workmanship shall be first class throughout. The laryngoscope blade assemblies, lamps, battery handle and case shall be free from defects which detract from their appearance or impair their serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality supplied by the manufacturer of the blades, handles, and lamps may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, the individual test results, and the specification requirements.

4.2 Sampling.

* 4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in table I. The unit of product for examination purposes shall be one complete laryngoscope consisting of all components specified in 3.2.

TABLE I. Sampling for examination.

Type of defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table II. Unit of product for test purposes shall be one complete laryngoscope consisting of all components specified in 3.2. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

TABLE II. Sampling for tests.

Component	Characteristic	Requirement	Test Procedure	Inspection level	AQL
Laryngoscope blade assemblies	Dimensions	3.2.1	---	S-1	2.5
	Internal thread test	3.2.1.1	4.4.1	S-2	2.5
	Plating test	3.1.1 and GG-I-526	GG-I-526	S-2	2.5
	Salt-spray test	3.1.1 and GG-I-526	GG-I-526	S-2	2.5
	Boil test	3.1.1 and GG-I-526	GG-I-526	S-2	2.5
	Hardness (For class 3 material)	3.1.1	4.4.2	S-2	2.5
Battery handle	For all requirements see MIL-L-36628.				
Lamps	For all requirements see MIL-L-36628 for Type II Lamps.				

* 4.2.3 Raw materials. A sufficient quantity of each lot of each metal used in the manufacture of the laryngoscopes shall be selected for chemical analysis.

* 4.3 Examination. The laryngoscopes shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted as specified in tables I and III. Examination, however, shall not be restricted to defects listed in table III. Any laryngoscopes in the sample containing one or more defects shall be rejected and, if the number of defective items in any sample exceeds the acceptance number for the sample, the lot represented by the sample shall be rejected.

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TABLE III. Classification of defects.

Categories	Defects
Major	
101	Lamps do not light when laryngoscope blade is inserted onto battery handle (when two "D" size batteries are inserted into battery handle).
102	Blades not of detachable, hook-on design with respect to battery handle.
103	Blades do not positively seat (hook-on) when inserted onto battery handle.
104	Blades not of finish specified.
105	Blades not free from sharp edges, crevices, burrs, grind marks or cracks.
106	Internal threads of lamp socket not complete or not well defined.
107	External threads of lamps not complete or not well defined.
108	Extra lamps not furnished with complete laryngoscope set.
109	Battery handle finish not bright.
110	Battery handle cannot accommodate two standard size "D" batteries.
111	Glass envelope of lamp not properly secured to lamp base.
112	Glass envelope of lamp cracked.
113	Plated parts not free from flaking.
114	Blades and battery handle not permanently or legibly marked as specified in 3.4.
Minor	
201	Battery handle inner surfaces not plated.
202	Glass envelope of lamp not free from bubbles, dirt or other imperfections.

* 4.3.2 Packaging inspection. The inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

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4.4.1 Internal thread test of lamp sockets. The contractor shall provide all the following thread plug gages to the Government Quality Assurance Representative (QAR) for checking the 8-32UNC-2B internal threads of the lamp sockets with respect to PITCH DIAMETER AND MINOR DIAMETER:

a. "GO" Thread Plug Gage for PITCH DIAMETER (X gage) Identification marking: 8-32 UNC GO PD .1437

b. "HI" Thread Plug Gage for PITCH DIAMETER (X gage) Identification marking: 8-32 UNC-2B HI PD .1475

c. "GO" Plain Plug Gage for MINOR DIAMETER (Z gage) Identification marking: 8-32 UNC GO .1300

d. "NOT GO" Plain Plug Gage for MINOR DIAMETER (Z gage) Identification marking: 8-32 UNC-2B NOT GO .1390

To perform thread tests of the lamp sockets, remove lamp from lamp socket and place plug gages into lamp sockets. The gaging procedure to be used, dimensions, tolerances and gage identifications shall be in accordance with ANSI Standard B 1.2. Fits of gages into internal lamp socket threads shall be noted.

4.4.2 Hardness test. Rockwell hardness of class 3 material shall be determined in accordance with ASTM E 18 or E 92. Conversion of Vickers hardness values to Rockwell values shall be in accordance with ASTM E 140.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.1).

5.1.1 Unit of issue. One laryngoscope as specified in 3.2 constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Cleaning. Each complete laryngoscope shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.2.2 Drying. Each complete laryngoscope shall be dried in accordance with MIL-P-116.

5.1.2.3 Preservation application. Preservatives shall not be used.

5.1.2.4 Unit packaging. Components comprising the complete laryngoscope along with extra lamps shall be wrapped in neutral tissue or other suitable non-corrosive material and individually packaged in a box of appropriate size and design. Each complete set shall be further packaged in accordance with MIL-P-116, method IC-2. The interior container shall be a corrugated box (see 3.2).

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5.1.3 Level C. Laryngoscopes shall be clean, dry and individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. Twenty-four unit packages shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.2 Level B. Twenty-four unit packages shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The packaged laryngoscopes shall be packed in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier's rules and regulations.

5.3 Marking.

5.3.1 Interior packages. Each unit package shall be marked as specified in MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Shipping container. Shipping containers shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) National Stock Number.

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6.2 This specification covers the following item appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6515-00-346-0480	LARYNGOSCOPE, Child-Adult, Wisconsin

6.3 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Offerors and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
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